

Ronish Shrestha

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Professional Summary

Machine Learning-focused Computer Science graduate with experience building predictive models and real-world data systems. Published IEEE research and developed an integrated AI platform combining health, environmental, and time-series data. Seeking to apply data science and ML to solve practical problems with measurable impact.

Education

Lamar University, Beaumont, TX Master of Science in Computer Science Expected December 2026

The British College (Leeds Beckett University, UK) Bachelor of Science in Computing, First Class Honours 2018 – 2022

Experience

Research Assistant Lamar University, Beaumont, TX Jan 2024 – Present

- Analyzed multi-source datasets (CDC, EPA, BRFSS, FEMA, NOAA) to identify patterns in environmental and health risk factors
- Built data pipelines using Python (Pandas, NumPy) for data cleaning, transformation, and feature engineering
- Performed spatial analysis (hotspot and cluster detection) using ArcGIS Pro
- Developed machine learning models (Random Forest, XGBoost) achieving 93
- Applied SHAP to interpret feature importance and improve model explainability
- Built time-series models (Random Forest, LSTM, ARIMA) using IPEDS and NSF HERD datasets
- Published research on machine learning-based KPI forecasting at IEEE SmartIoT 2025
- Built an AI system integrating image, time-series, and tabular models using FastAPI and Docker (journal under review)

Data Specialist CloudFactory, Kathmandu, Nepal July 2021 – July 2024

- Processed and validated large-scale structured and geospatial datasets
- Improved data quality through cleaning, transformation, and validation processes
- Created visualizations to communicate data insights for analysis and decision-making
- Streamlined workflows to improve data processing efficiency

Projects

AI Data Lakehouse for Health Vulnerability Analysis

- Integrated CDC PLACES, EPA EJScreen, and BRFSS datasets for combined analysis
- Built predictive models (XGBoost, Random Forest)
- Applied SHAP to analyze feature contributions across multiple domains

University KPI Forecasting (Published – IEEE SmartIoT 2025)

- Built forecasting models for enrollment, employees, graduates, and R&D expenditure
- Evaluated models using MAE, RMSE, MAPE, and R^2
- Identified Random Forest as more stable than LSTM for short time-series data

Centralized AI Lakehouse Framework (Journal Under Review)

- Built an end-to-end system integrating MRI, KPI, and water quality models
- Developed REST APIs using FastAPI
- Deployed using Docker for reproducibility and scalability

Publications

IEEE SmartIoT 2025 Scalable and Robust AI Forecasting of University KPIs Using LSTM and Random Forest

Under Review Centralized AI Lakehouse Framework for Brain Tumor Analysis, KPI Forecasting, and Water Quality Prediction

Technical Skills

Programming: Python, JavaScript, SQL

Data Analysis: Pandas, NumPy, Data Cleaning, Feature Engineering

Machine Learning: Scikit-learn, XGBoost, Random Forest

Time Series: LSTM, ARIMA

Explainability: SHAP

Deep Learning: PyTorch, CNN, ResNet50, DeepLabV3

APIs & Deployment: FastAPI, REST APIs, Docker

Visualization: Matplotlib

GIS: ArcGIS Pro

Tools: Git, GitHub, LaTeX